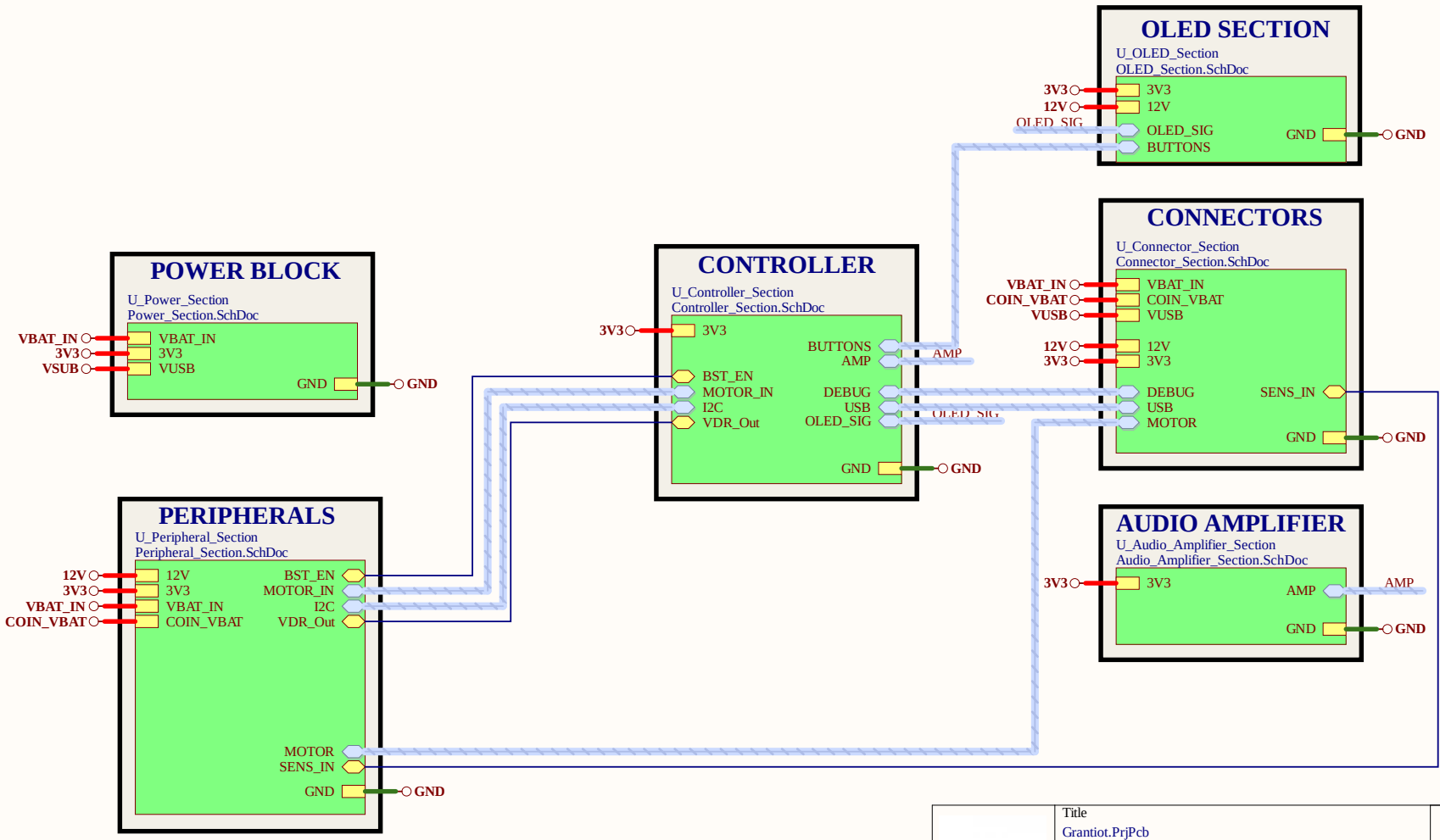
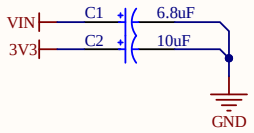
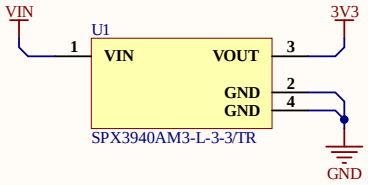


Title	Grantiot.PrjPcb		
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A4	1		1
Date:	3/30/2022	Sheet 1 of	9
File:	E:\Git\..\Top Sheet.SchDoc	Drawn By:	Huzaifa Ali

DECOUPLING CAPS



MCU VOLTAGE REGULATOR IC



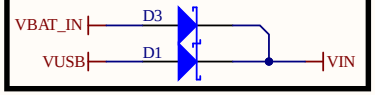
TEST POINT



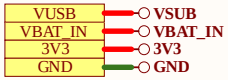
POWER INDICATION LED



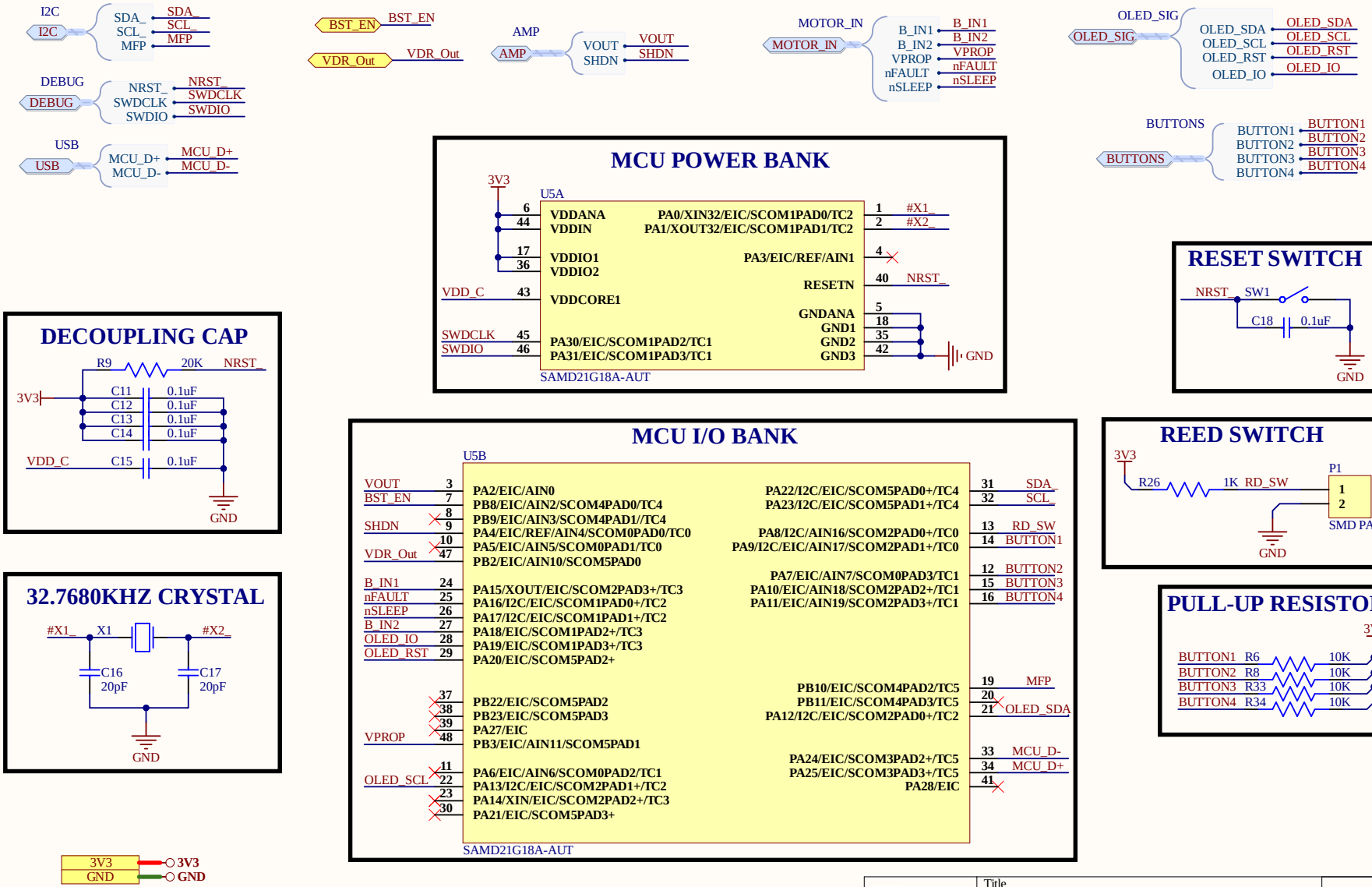
ORING CIRCUIT



VBAT=6V
VUSB=5V
VF of Diode= 360mV @1A
VR=30V
IF=1A



Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/30/2022		Sheet 2 of 9	
File: E:\Git\..\Power_Section.SchDoc		Drawn By: Huzaifa Ali	



Controller Section

Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/30/2022	Sheet 5 of	9
File:	E:\Git\..\Controller_Section.SchDoc	Drawn By:	Huzaifa Ali

BST_EN BST_EN

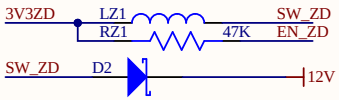
I2C
I2C SDA_ SDA_
SCL_ SCL_
MFP MFP

MOTOR
MOTOR OUT_1 OUT_1
OUT_2 OUT_2

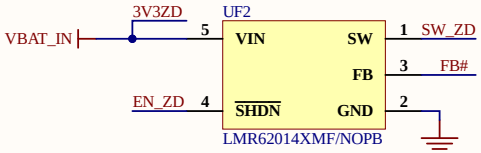
MOTOR_IN
MOTOR_IN B_IN1 B_IN1
B_IN2 B_IN2
VPROP VPROP
nFAULT nFAULT
nSLEEP nSLEEP

SENS_IN SENS_IN
VDR_Out VDR_Out

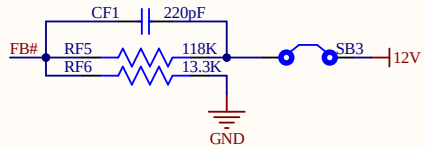
SUPPORTING CIRCUIT



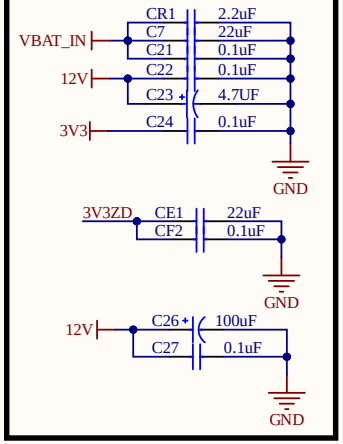
12V BOOST REGULATOR



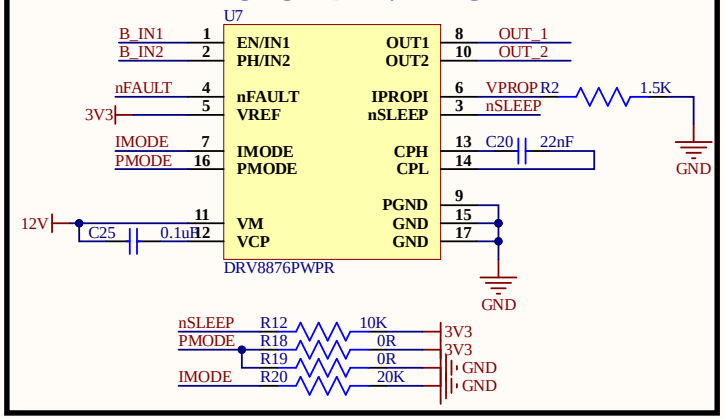
FEEDBACK NETWORK



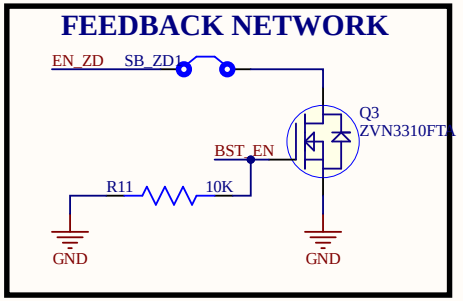
DECOUPLING CAPS



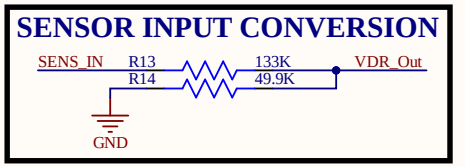
MOTOR DRIVER IC



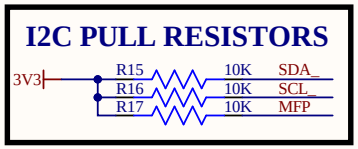
FEEDBACK NETWORK



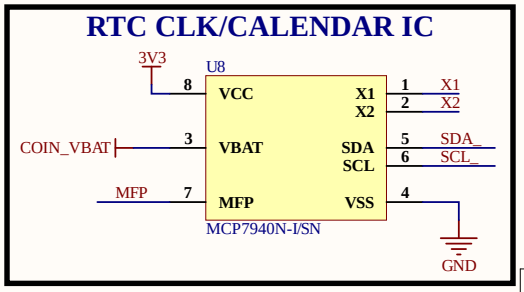
SENSOR INPUT CONVERSION



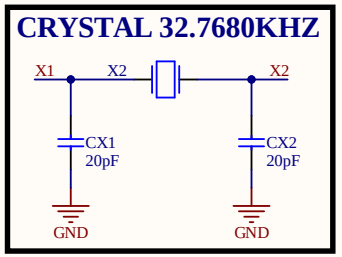
I2C PULL RESISTORS



RTC CLK/CALENDAR IC



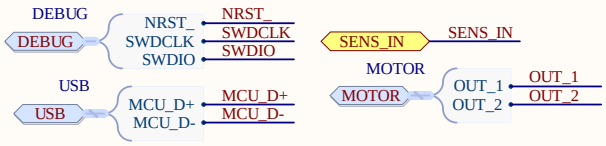
CRYSTAL 32.7680KHZ



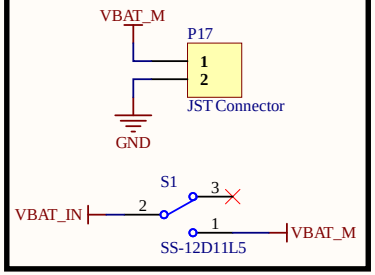
COIN_VBAT — COIN_VBAT
VBAT_IN — VBAT_IN
12V — 12V
3V3 — 3V3
GND — GND

Peripheral Section

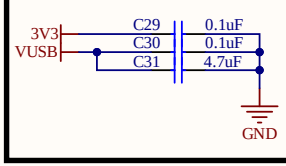
Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
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File: E:\Git\..\Peripheral_Section.SchDoc		Drawn By: Huzaifa Ali	



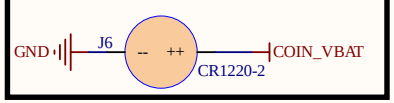
BATTERY INPUT



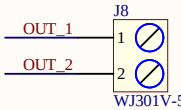
DECOUPLING CAP



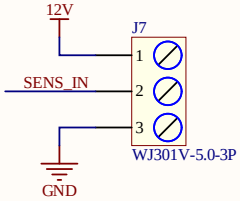
COIN CELL CONNECTOR



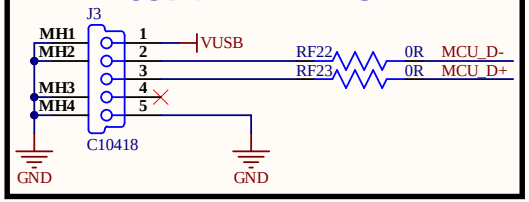
MOTOR OUTPUT



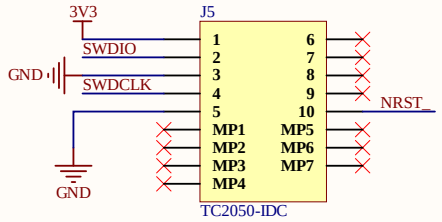
PROXIMITY SENSOR



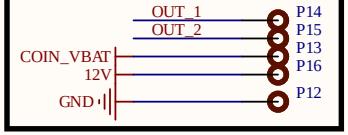
USB DATA TRANSFER



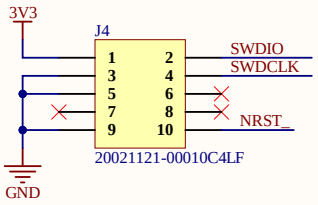
JTAG DEBUGGING CONNECTOR

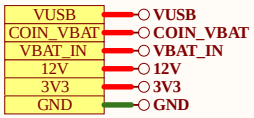


TEST POINTS



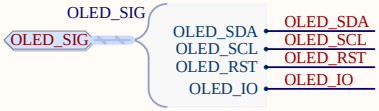
SERIAL WIRE DEBUGGING HEADER



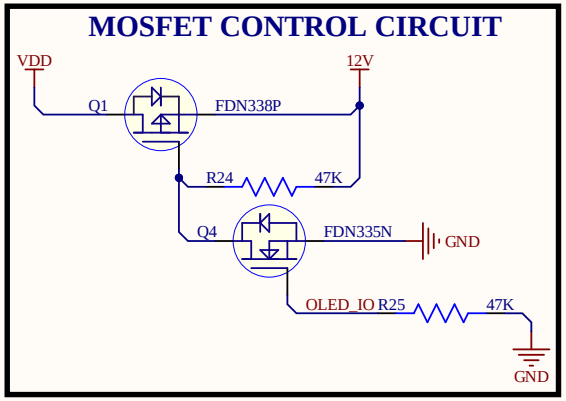


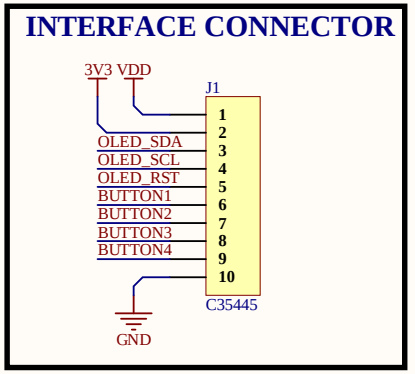
Connector Section

Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/30/2022		Sheet 7 of 9	
File: E:\Git\..\Connector_Section.SchDoc		Drawn By: Huzaifa Ali	











Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/30/2022		Sheet 8 of 9	
File: E:\Git\..\OLED_Section.SchDoc		Drawn By: Huzaifa Ali	

AMP

AMP

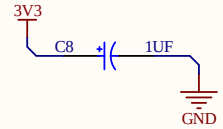
VOUT

SHDN

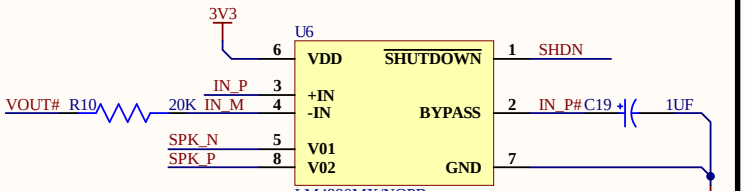
VOUT

SHDN

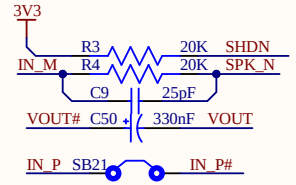
DECOUPLING CAP



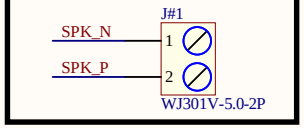
1W AUDIO AMPLIFIER IC



SUPPORTING CIRCUIT



SPEAKER OUTPUT



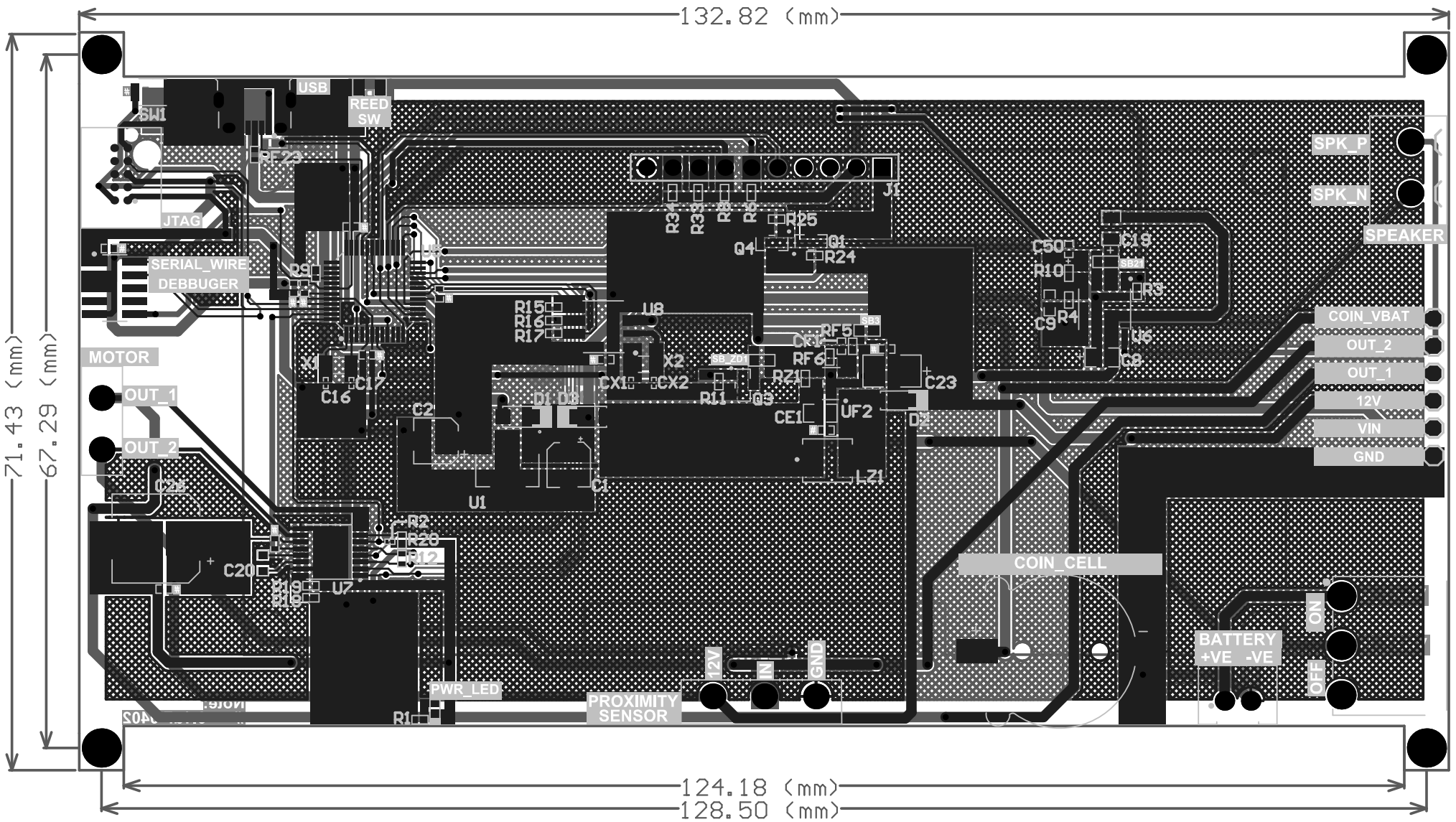
3V3

GND

3V3

GND

Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/30/2022		Sheet 9 of 9	
File: E:\Git\..\Audio_Amplifier_Section.SchDoc		Drawn By: Huzaifa Ali	


132.82 (mm)
71.43 (mm)
67.29 (mm)
124.18 (mm)
128.50 (mm)
USB
REED SW
SW1
RF23
JTAG
SERIAL_WIRE DEBBUGER
MOTOR
OUT_1
OUT_2
SPK_P
SPK_N
SPEAKER
COIN_VBAT
OUT_2
OUT_1
12V
VIN
GND
COIN_CELL
BATTERY +VE -VE
ON
OFF
PWR_LED
PROXIMITY SENSOR
12V
IN
GND
J1
U1
U2
U6
U7
U8
C1
C2
C9
C16
C17
C19
C20
C23
C50
R1
R2
R9
R10
R15
R16
R17
R24
R25
Q1
Q4
D1
D2
D3
LZ1
CE1
UF2
RF5
RF6
RZ1
CX1
CX2
X1
X2